IMPERIAL

Department of Computing
70015: Mathematics for Machine Learning
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Practice Problems 02: Generalized Linear Models & Vector Calculus

Basic Vector Calculus

Question 27 (Basic vector derivatives). (I)
Let $\mathbf{u}, \mathbf{v}, \boldsymbol{\theta} \in \mathbb{R}^n$ be column vectors, and let $\mathbf{A} \in \mathbb{R}^{n \times n}$.

a. For $f(\mathbf{v}) = \mathbf{u}^\top \mathbf{v}$, find $\frac{\partial f}{\partial \mathbf{v}}$.

b. For $f(\mathbf{u}) = \mathbf{u}^\top \mathbf{v}$, find $\frac{\partial f}{\partial \mathbf{u}}$.

c. For $\mathbf{f}(\mathbf{v}) = \mathbf{A}\mathbf{v}$, find $\frac{\partial \mathbf{f}}{\partial \mathbf{v}}$.

d. For $\mathbf{f}(\mathbf{v}) = \mathbf{v}^\top \mathbf{A}$, find $\frac{\partial \mathbf{f}}{\partial \mathbf{v}}$.

e. For $f(\boldsymbol{\theta}) = \boldsymbol{\theta}^\top \mathbf{A}\boldsymbol{\theta}$, find $\frac{\partial f}{\partial \boldsymbol{\theta}}$.

Linear Regression

Question 28 (Simplification of the Quadratic Form Gradient). (I)
In lecture, we derived the identity:

$$\nabla_\theta(\theta^\top A\theta) = (A + A^\top)\theta.$$

In this exercise, assume that A is symmetric, i.e., $A = A^\top$, and show that the above gradient simplifies to $2A\theta$. *Hint: this is also a nice opportunity to practice index notation skills.*

Question 29 (Ridge Regression Estimator). (II)
In our lecture we saw the ordinary least squares estimator, but discussed how regularization may be needed to smooth our model's outputs. Given the below ridge regression loss, please derive the optimal parameter.

$$\mathcal{L}_{\text{ridge}}(\theta) := \frac{1}{k}||\mathbf{y} - \boldsymbol{X}\theta||^2 + \lambda||\theta||_2^2$$

Question 30 (Legendre Basis Expansion and Orthogonality). IV

In lecture, we discussed how polynomial basis expansions can be used to extend linear models to nonlinear relationships. However, we also saw that high-degree monomial bases (e.g., $[1, x, x^2, x^3, \ldots]$) can lead to highly correlated features and unstable estimates.

(a) Using Wikipedia, write down the first four Legendre polynomials $P_0(x)$ through $P_3(x)$. Then, define a basis expansion function $\phi(x)$ that maps a scalar input $x \in \mathbb{R}$ to a vector of its Legendre basis terms up to degree 3.

(b) Legendre polynomials are known to be orthogonal on the interval $[-1, 1]$. Assuming we use these polynomials as columns in our design matrix $\boldsymbol{X}$, what does this imply about the structure of $\boldsymbol{X}^\top \boldsymbol{X}$ in our ordinary least squares formulation? (You do not need to prove anything formally, simply reason about the effects.)

(c) Implement this Legendre basis expansion in Python (hint: use `numpy.polynomial.legendre.legvander`) and compare the condition number of $\boldsymbol{X}^\top \boldsymbol{X}$ to that obtained when using the standard monomial basis. What do you observe?

Topical Applications

Question 31 (Gradient of the Logistic Regression Loss). II

In logistic regression, we model the confidence of class 1 as

$$\hat{\boldsymbol{y}} = \sigma(\boldsymbol{X}\theta), \qquad \sigma(z) = \frac{1}{1 + e^{-z}},$$

where $\boldsymbol{X} \in \mathbb{R}^{m \times d}$ is the design matrix containing m examples and $\theta \in \mathbb{R}^d$ are the model parameters.

The negative log-likelihood (empirical risk) of this model can be written compactly as

$$\mathcal{L}(\theta) = -\frac{1}{m}\Big[\boldsymbol{y}^\top \log(\sigma(\boldsymbol{X}\theta)) + (\mathbf{1} - \boldsymbol{y})^\top \log(1 - \sigma(\boldsymbol{X}\theta))\Big].$$

Derive the gradient of $\mathcal{L}(\theta)$ with respect to θ.

Question 32 (Bradley–Terry Model and Preference Learning). III

Pairwise comparison models such as the **Bradley–Terry model** are widely used for *preference modeling*—for example, in large language model (LLM) alignment, where human annotators express which of two responses is preferred. Each response (or player) i is assigned a latent *score* parameter β_i, reflecting its underlying quality or skill.

We represent a comparison between two responses i and j using a feature vector $\boldsymbol{x}_{ij} \in \mathbb{R}^n$ defined as:

$$(\boldsymbol{x}_{ij})_k = \begin{cases} +1, & \text{if } k = i, \\ -1, & \text{if } k = j, \\ 0, & \text{otherwise.} \end{cases}$$

Collecting all such feature vectors into a matrix $\boldsymbol{X}$, we also record the outcomes in a label vector $\boldsymbol{y}$, where $y_{ij} = 1$ if i beats j, and 0 otherwise.

(a) Applying a linear model to $\boldsymbol{x}_{ij}$ gives a scalar value $\boldsymbol{x}_{ij}^\top \boldsymbol{\beta}$. Express this term in simpler form, using index notation if helpful. What does it represent intuitively?

(b) To model the probability that i beats j as a classification problem, which nonlinearity from our notes should we apply to $\boldsymbol{x}_{ij}^\top \boldsymbol{\beta}$? Using your answer to part (a), write this probability explicitly.

(c) Derive the gradient of the log-likelihood with respect to $\boldsymbol{\beta}$ for this model.